

$O(n \log n)$ Time Complexity

$O(n \log n)$ is a time complexity where:

The input size is n

The algorithm performs $\log n$ operations for each element

So total work = $n \times \log n$

1. Merge Sort

```
def merge_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    mid = len(arr) // 2  
    left = merge_sort(arr[:mid])  
    right = merge_sort(arr[mid:])  
  
    return merge(left, right)
```

2. Quick Sort (average case)

3. Heap Sort